

Figure 7-4 A transverse electromagnetic (TEM) wave propagating in the direction $\hat{\mathbf{k}} = \hat{\mathbf{z}}$. For all TEM waves, $\hat{\mathbf{k}}$ is parallel to $\mathbf{E} \times \mathbf{H}$.

and

$$\mathbf{H}(z,t) = \Re\mathfrak{e}\left[\widetilde{\mathbf{H}}(z)\, e^{j\omega t}\right] = \hat{\mathbf{y}}\frac{|E_{x0}^{+}|}{\eta}\cos(\omega t - kz + \phi^{+}) \quad \text{(A/m)}. \tag{7.34b}$$

Because $\mathbf{E}(z,t)$ and $\mathbf{H}(z,t)$ exhibit the same functional dependence on z and t, they are said to be ***in phase***; when the amplitude of one of them reaches a maximum, the amplitude of the other does so too. The fact that $\widetilde{\mathbf{E}}$ and $\widetilde{\mathbf{H}}$ are in phase is characteristic of waves propagating in lossless media.

From the material on wave motion presented in Section 1-4, we deduce that the ***phase velocity*** of the wave is

$$u_p = \frac{\omega}{k} = \frac{\omega}{\omega\sqrt{\mu\varepsilon}} = \frac{1}{\sqrt{\mu\varepsilon}} \quad \text{(m/s)}, \tag{7.35}$$

and its wavelength is

$$\lambda = \frac{2\pi}{k} = \frac{u_p}{f} \quad \text{(m)}. \tag{7.36}$$

In a vacuum, $\varepsilon = \varepsilon_0$ and $\mu = \mu_0$, and the phase velocity u_p and the intrinsic impedance η given by Eq. (7.31) are

$$u_p = c = \frac{1}{\sqrt{\mu_0\varepsilon_0}} = 3 \times 10^8 \quad \text{(m/s)}, \tag{7.37}$$

$$\eta = \eta_0 = \sqrt{\frac{\mu_0}{\varepsilon_0}} = 377\ (\Omega) \approx 120\pi \quad (\Omega), \tag{7.38}$$

where c is the velocity of light and η_0 is called the ***intrinsic impedance of free space***.

Example 7-1: EM Plane Wave in Air

This example is analogous to the "Sound Wave in Water" problem given by Example 1-1.

The electric field of a 1 MHz plane wave traveling in the $+z$ direction in air points along the x direction. If this field reaches a peak value of 1.2π (mV/m) at $t = 0$ and $z = 50$ m, obtain expressions for $\mathbf{E}(z,t)$ and $\mathbf{H}(z,t)$. Then plot them as a function of z at $t = 0$.

Solution: At $f = 1$ MHz, the wavelength in air is

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{1 \times 10^6} = 300 \text{ m},$$

and the corresponding wavenumber is $k = (2\pi/300)$ (rad/m). The general expression for an x-directed electric field traveling in the $+z$ direction is given by Eq. (7.34a) as

$$\mathbf{E}(z,t) = \hat{\mathbf{x}}|E_{x0}^{+}|\cos(\omega t - kz + \phi^{+})$$
$$= \hat{\mathbf{x}}\,1.2\pi\cos\left(2\pi \times 10^6 t - \frac{2\pi z}{300} + \phi^{+}\right) \quad \text{(mV/m)}.$$

The field $\mathbf{E}(z,t)$ is maximum when the argument of the cosine function equals zero or a multiple of 2π. At $t = 0$ and $z = 50$ m, this condition yields

$$-\frac{2\pi \times 50}{300} + \phi^{+} = 0 \qquad \text{or} \qquad \phi^{+} = \frac{\pi}{3}.$$

Hence,

$$\mathbf{E}(z,t) = \hat{\mathbf{x}}\,1.2\pi\cos\left(2\pi \times 10^6 t - \frac{2\pi z}{300} + \frac{\pi}{3}\right) \quad \text{(mV/m)},$$

and from Eq. (7.34b), we have

$$\mathbf{H}(z,t) = \hat{\mathbf{y}}\frac{E(z,t)}{\eta_0}$$
$$= \hat{\mathbf{y}}\,10\cos\left(2\pi \times 10^6 t - \frac{2\pi z}{300} + \frac{\pi}{3}\right) \quad (\mu\text{A/m}),$$

where we have used the approximation $\eta_0 \approx 120\pi\ (\Omega)$.

At $t = 0$,

$$\mathbf{E}(z,0) = \hat{\mathbf{x}}\,1.2\pi\cos\left(\frac{2\pi z}{300} - \frac{\pi}{3}\right) \quad \text{(mV/m)},$$

$$\mathbf{H}(z,0) = \hat{\mathbf{y}}\,10\cos\left(\frac{2\pi z}{300} - \frac{\pi}{3}\right) \quad (\mu\text{A/m}).$$

Plots of $\mathbf{E}(z,0)$ and $\mathbf{H}(z,0)$ as a function of z are shown in **Fig. 7-5**.

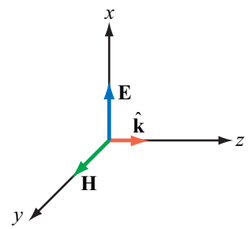


图 7-4 沿 k̂ = ẑ 方向传播的横向电磁 (TEM) 波。对于所有 TEM 波，k̂ 平行于 E× H。

和

$$\mathbf{H}(z,t) = \Re\mathfrak{e}\left[\widetilde{\mathbf{H}}(z)\, e^{j\omega t}\right] = \hat{\mathbf{y}}\frac{|E_{x0}^{+}|}{\eta}\cos(\omega t - kz + \phi^{+}) \quad \text{(A/m)}. \tag{7.34b}$$

因为 E(z,t) 和 H(z,t) 对 z 和 t 表现出相同的函数依赖性，所以称它们同相；当其中一个的幅度达到最大值时，另一个的幅度也会达到最大值。 ~E 和 ~H 同相这一事实是波在无损介质中传播的特征。

根据第 1-4 节中介绍的有关波动的材料，我们推断出波的相速度为

$$u_p = \frac{\omega}{k} = \frac{\omega}{\omega\sqrt{\mu\varepsilon}} = \frac{1}{\sqrt{\mu\varepsilon}} \quad \text{(m/s)}, \tag{7.35}$$

其波长为

$$\lambda = \frac{2\pi}{k} = \frac{u_p}{f} \quad \text{(m)}. \tag{7.36}$$

在真空中，ε = ε0和μ = μ0，以及由方程给出的相速度up和固有阻抗η。 (7.31) 是

$$u_p = c = \frac{1}{\sqrt{\mu_0\varepsilon_0}} = 3 \times 10^8 \quad \text{(m/s)}, \tag{7.37}$$

$$\eta = \eta_0 = \sqrt{\frac{\mu_0}{\varepsilon_0}} = 377\ (\Omega) \approx 120\pi \quad (\Omega), \tag{7.38}$$

其中 c 是光速，η_0 称为自由空间的本征阻抗。

例 7-1：空气中的电磁平面波

此示例类似于示例 1-1 中给出的 "水中的声波" 问题。

空气中沿 +z 方向传播的 1 MHz 平面波的电场沿 x 方向指向。如果该场在 t = 0 和 z = 50 m 处达到峰值 1.2π (mV/m)，则获得 E(z,t) 和 H(z,t) 的表达式。然后将它们绘制为 t = 0 处 z 的函数。

解： 在 f = 1 MHz 时，空气中的波长为

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{1 \times 10^6} = 300 \text{ m},$$

对应的波数为k = (2π/300) (rad/m)。沿 +z 方向传播的 x 方向电场的一般表达式由方程 1 给出。 (7.34a) 为

$$\mathbf{E}(z,t) = \hat{\mathbf{x}}|E_{x0}^{+}|\cos(\omega t - kz + \phi^{+})$$
$$= \hat{\mathbf{x}}\,1.2\pi\cos\left(2\pi \times 10^6 t - \frac{2\pi z}{300} + \phi^{+}\right) \quad \text{(mV/m)}.$$

当余弦函数的参数等于零或 2π 的倍数时，域 E(z,t) 最大。在 t = 0 和 z = 50 m 处，此条件产生

$$-\frac{2\pi \times 50}{300} + \phi^{+} = 0 \qquad \text{or} \qquad \phi^{+} = \frac{\pi}{3}.$$

因此，

$$\mathbf{E}(z,t) = \hat{\mathbf{x}}\,1.2\pi\cos\left(2\pi \times 10^6 t - \frac{2\pi z}{300} + \frac{\pi}{3}\right) \quad \text{(mV/m)},$$

并从方程式(7.34b)，我们有

$$\mathbf{H}(z,t) = \hat{\mathbf{y}}\frac{E(z,t)}{\eta_0}$$
$$= \hat{\mathbf{y}}\,10\cos\left(2\pi \times 10^6 t - \frac{2\pi z}{300} + \frac{\pi}{3}\right) \quad (\mu\text{A/m}),$$

我们使用了近似值 $\eta_0 \approx 120\pi\ (\Omega)$。

在 t = 0 时，

$$\mathbf{E}(z,0) = \hat{\mathbf{x}}\,1.2\pi\cos\left(\frac{2\pi z}{300} - \frac{\pi}{3}\right) \quad \text{(mV/m)},$$

$$\mathbf{H}(z,0) = \hat{\mathbf{y}}\,10\cos\left(\frac{2\pi z}{300} - \frac{\pi}{3}\right) \quad (\mu\text{A/m}).$$

E(z, 0) 和 H(z, 0) 作为 z 函数的图如图 7-5 所示。

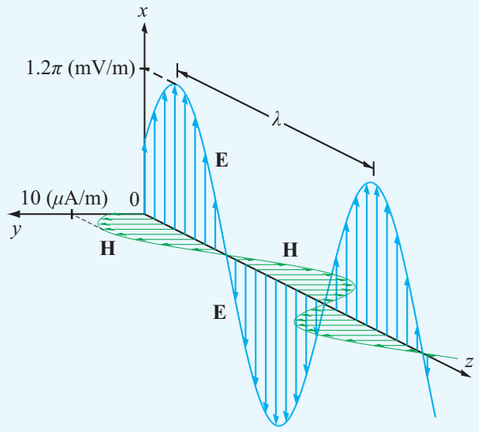


Figure 7-5 Spatial variations of **E** and **H** at $t = 0$ for the plane wave of Example 7-1.

Wave Propagating Along $+z$ with E Along $\hat{\mathbf{x}}$

Let us apply Eq. (7.39a) to the wave given by Eq. (7.32a). The direction of propagation $\hat{\mathbf{k}} = \hat{\mathbf{z}}$ and $\widetilde{\mathbf{E}} = \hat{\mathbf{x}}\widetilde{E}_x^+(z)$. Hence,

$$\widetilde{\mathbf{H}} = \frac{1}{\eta}\hat{\mathbf{k}} \times \widetilde{\mathbf{E}} = \frac{1}{\eta}(\hat{\mathbf{z}} \times \hat{\mathbf{x}})\widetilde{E}_x^+(z) = \hat{\mathbf{y}}\frac{\widetilde{E}_x^+(z)}{\eta}, \tag{7.40}$$

which is the same as the result given by Eq. (7.32b).

Wave Propagating Along $-z$ with E Along $\hat{\mathbf{x}}$

For a wave traveling in the $-z$ direction with electric field given by

$$\widetilde{\mathbf{E}} = \hat{\mathbf{x}}\widetilde{E}_x^-(z) = \hat{\mathbf{x}}E_{x0}^- e^{jkz}, \tag{7.41}$$

application of Eq. (7.39a) gives

$$\widetilde{\mathbf{H}} = \frac{1}{\eta}(-\hat{\mathbf{z}} \times \hat{\mathbf{x}})\widetilde{E}_x^-(z) = -\hat{\mathbf{y}}\frac{\widetilde{E}_x^-(z)}{\eta} = -\hat{\mathbf{y}}\frac{E_{x0}^-}{\eta}e^{jkz}. \tag{7.42}$$

Hence, in this case, $\widetilde{\mathbf{H}}$ points in the negative y direction.

7-2.2 General Relation between E and H

It can be shown that, for any uniform plane wave traveling in an arbitrary direction denoted by the unit vector $\hat{\mathbf{k}}$, the electric and magnetic field phasors $\widetilde{\mathbf{E}}$ and $\widetilde{\mathbf{H}}$ are related as

$$\widetilde{\mathbf{H}} = \frac{1}{\eta}\hat{\mathbf{k}} \times \widetilde{\mathbf{E}}, \tag{7.39a}$$

$$\widetilde{\mathbf{E}} = -\eta\hat{\mathbf{k}} \times \widetilde{\mathbf{H}}. \tag{7.39b}$$

► The following right-hand rule applies: When we rotate the four fingers of the right hand from the direction of **E** toward that of **H**, the thumb points in the direction of the wave travel, $\hat{\mathbf{k}}$. ◄

The relations given by Eqs. (7.39a and b) are valid not only for lossless media but for lossy ones as well. As we see later in Section 7-4, the expression for η of a lossy medium is different from that given by Eq. (7.31). As long as the expression used for η is appropriate for the medium in which the wave is traveling, the relations given by Eqs. (7.39a and b) always hold.

Wave Propagating Along $+z$ with E Along $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$

In general, a uniform plane wave traveling in the $+z$ direction may have both x and y components, where $\widetilde{\mathbf{E}}$ is given by

$$\widetilde{\mathbf{E}} = \hat{\mathbf{x}}\widetilde{E}_x^+(z) + \hat{\mathbf{y}}\widetilde{E}_y^+(z), \tag{7.43a}$$

and the associated magnetic field is

$$\widetilde{\mathbf{H}} = \hat{\mathbf{x}}\widetilde{H}_x^+(z) + \hat{\mathbf{y}}\widetilde{H}_y^+(z). \tag{7.43b}$$

Application of Eq. (7.39a) gives

$$\widetilde{\mathbf{H}} = \frac{1}{\eta}\hat{\mathbf{z}} \times \widetilde{\mathbf{E}} = -\hat{\mathbf{x}}\frac{\widetilde{E}_y^+(z)}{\eta} + \hat{\mathbf{y}}\frac{\widetilde{E}_x^+(z)}{\eta}. \tag{7.44}$$

By equating Eq. (7.43b) to Eq. (7.44), we have

$$\widetilde{H}_x^+(z) = -\frac{\widetilde{E}_y^+(z)}{\eta}, \qquad \widetilde{H}_y^+(z) = \frac{\widetilde{E}_x^+(z)}{\eta}. \tag{7.45}$$

These results are illustrated in **Fig. 7-6**. The wave may be considered the sum of two waves: one with electric and magnetic components (E_x^+, H_y^+) and another with components (E_y^+, H_x^+). In general, a TEM wave may have an electric field in any direction in the plane orthogonal to the direction of wave travel, the associated magnetic field is also in the same plane, and its direction is dictated by Eq. (7.39a).

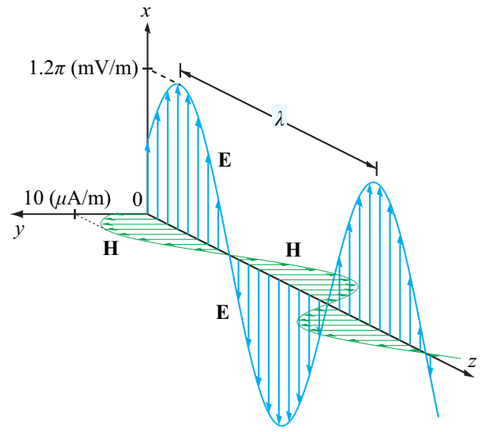


图 7-5 例 7-1 的平面波在 t = 0 时 E 和 H 的空间变化。

波沿 +z 传播，E 沿 x̂ 传播

让我们应用等式。 (7.39a) 到方程给出的波。 (7.32a) 。传播方向 $\hat{\mathbf{k}} = \hat{\mathbf{z}}$ 和 $\widetilde{\mathbf{E}} = \hat{\mathbf{x}}\widetilde{E}_x^+(z)$。因此，

$$\widetilde{\mathbf{H}} = \frac{1}{\eta}\hat{\mathbf{k}} \times \widetilde{\mathbf{E}} = \frac{1}{\eta}(\hat{\mathbf{z}} \times \hat{\mathbf{x}})\widetilde{E}_x^+(z) = \hat{\mathbf{y}}\frac{\widetilde{E}_x^+(z)}{\eta}, \tag{7.40}$$

这与等式给出的结果相同。 (7.32b)。

波沿 −z 传播，E 沿 x̂ 传播

对于沿 −z 方向传播的波，其电场由下式给出

$$\widetilde{\mathbf{E}} = \hat{\mathbf{x}}\widetilde{E}_x^-(z) = \hat{\mathbf{x}}E_{x0}^- e^{jkz}, \tag{7.41}$$

方程的应用(7.39a) 给出

$$\widetilde{\mathbf{H}} = \frac{1}{\eta}(-\hat{\mathbf{z}} \times \hat{\mathbf{x}})\widetilde{E}_x^-(z) = -\hat{\mathbf{y}}\frac{\widetilde{E}_x^-(z)}{\eta} = -\hat{\mathbf{y}}\frac{E_{x0}^-}{\eta}e^{jkz}. \tag{7.42}$$

因此，在这种情况下，$\widetilde{\mathbf{H}}$ 指向负 y 方向。

7-2.2 E与H的一般关系

可以证明，对于以单位矢量 $\hat{\mathbf{k}}$ 表示的任意方向传播的任何均匀平面波，电场相量 $\widetilde{\mathbf{E}}$ 和 $\widetilde{\mathbf{H}}$ 的关系为

$$\widetilde{\mathbf{H}} = \frac{1}{\eta}\hat{\mathbf{k}} \times \widetilde{\mathbf{E}}, \tag{7.39a}$$

$$\widetilde{\mathbf{E}} = -\eta\hat{\mathbf{k}} \times \widetilde{\mathbf{H}}. \tag{7.39b}$$

► 以下右手定则适用：当我们将右手的四个手指从 E 方向向 H 方向旋转时，拇指指向波传播的方向 $\hat{\mathbf{k}}$。 ◄

等式给出的关系。 (7.39a和b)不仅对于无损媒体有效，而且对于有损媒体也有效。正如我们稍后在第 7-4 节中看到的，有损介质的 η 的表达式与等式 1 给出的不同。 (7.31) 。只要用于 η 的表达式适用于波传播的介质，方程 1 给出的关系就可以得到。 (7.39a 和 b) 始终成立。

波沿着 +z 传播，E 沿着 x̂ 和 ŷ

一般来说，沿 +z 方向传播的均匀平面波可能同时具有 x 和 y 分量，其中 $\widetilde{\mathbf{E}}$ 由下式给出

$$\widetilde{\mathbf{E}} = \hat{\mathbf{x}}\widetilde{E}_x^+(z) + \hat{\mathbf{y}}\widetilde{E}_y^+(z), \tag{7.43a}$$

相关的磁场是

$$\widetilde{\mathbf{H}} = \hat{\mathbf{x}}\widetilde{H}_x^+(z) + \hat{\mathbf{y}}\widetilde{H}_y^+(z). \tag{7.43b}$$

方程的应用(7.39a) 给出

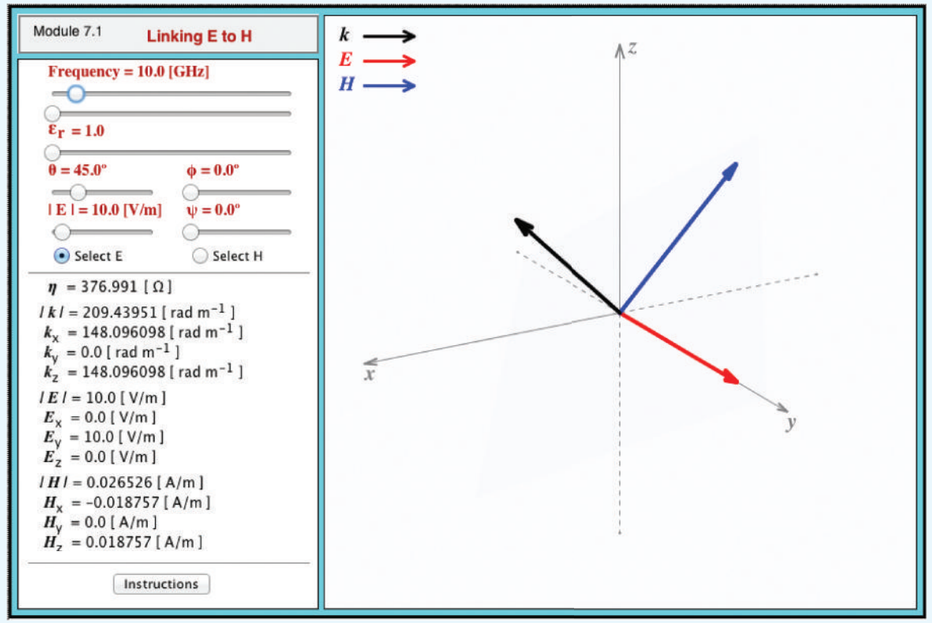
$$\widetilde{\mathbf{H}} = \frac{1}{\eta}\hat{\mathbf{z}} \times \widetilde{\mathbf{E}} = -\hat{\mathbf{x}}\frac{\widetilde{E}_y^+(z)}{\eta} + \hat{\mathbf{y}}\frac{\widetilde{E}_x^+(z)}{\eta}. \tag{7.44}$$

通过等式。 (7.43b) 至方程式（7.44），我们有

$$\widetilde{H}_x^+(z) = -\frac{\widetilde{E}_y^+(z)}{\eta}, \qquad \widetilde{H}_y^+(z) = \frac{\widetilde{E}_x^+(z)}{\eta}. \tag{7.45}$$

这些结果如图 7-6 所示。该波可以被认为是两个波的总和：一个具有电磁分量（E_x^+，H_y^+），另一个具有分量（E_y^+，H_x^+）。一般来说，TEM 波可以在与波传播方向正交的平面内的任何方向上产生电场，相关的磁场也在同一平面内，其方向由方程 1 决定。 (7.39a) 。

Module 7.1 Linking E to H Select the directions and magnitudes of **E** and **H** and observe the resultant wave vector.



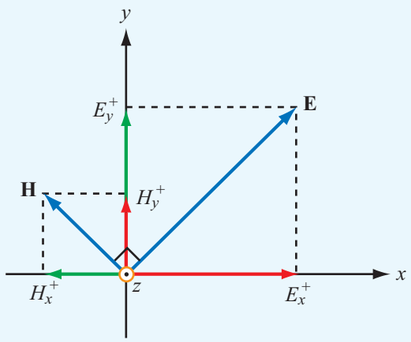


Figure 7-6 The wave $(\mathbf{E},\mathbf{H})$ is equivalent to the sum of two waves: one with fields (E_x^+,H_y^+) and another with (E_y^+,H_x^+) with both traveling in the $+z$ direction.

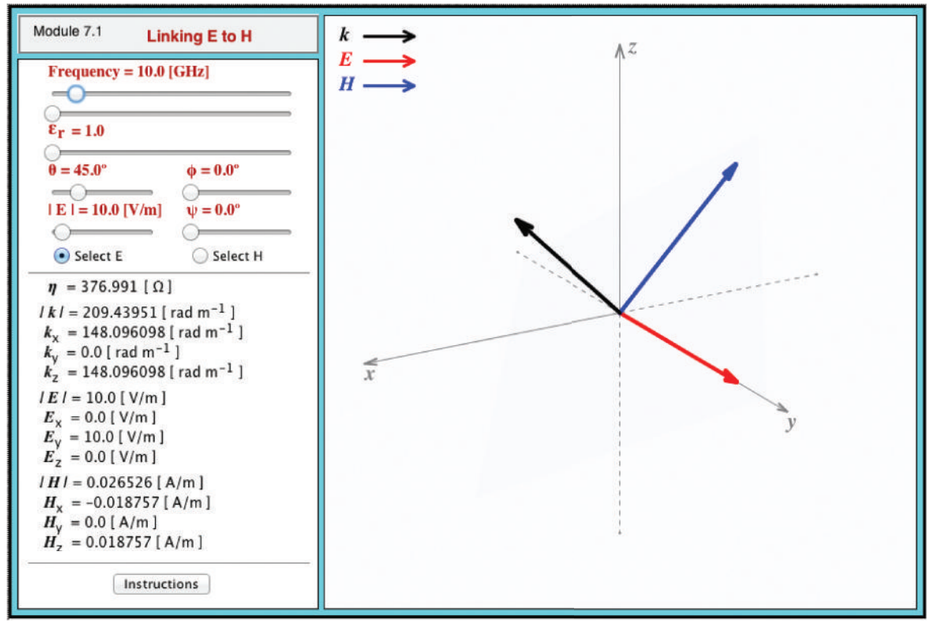
Concept Question 7-1: What is a uniform plane wave? Describe its properties, both physically and mathematically. Under what conditions is it appropriate to treat a spherical wave as a plane wave?

Concept Question 7-2: Since $\widetilde{\mathbf{E}}$ and $\widetilde{\mathbf{H}}$ are governed by wave equations of the same form [Eqs. (7.15) and (7.16)], does it follow that $\widetilde{\mathbf{E}} = \widetilde{\mathbf{H}}$? Explain.

Concept Question 7-3: If a TEM wave is traveling in the $\hat{\mathbf{y}}$ direction, can its electric field have components along $\hat{\mathbf{x}}$, $\hat{\mathbf{y}}$, and $\hat{\mathbf{z}}$? Explain.

Exercise 7-1: A 10-MHz uniform plane wave is traveling in a nonmagnetic medium with $\mu = \mu_0$ and $\varepsilon_r = 9$. Find (a) the phase velocity, (b) the wavenumber, (c) the wavelength in the medium, and (d) the intrinsic impedance of the medium.

模块 7.1 将 E 链接到 H 选择 E 和 H 的方向和大小并观察合成的波矢量。



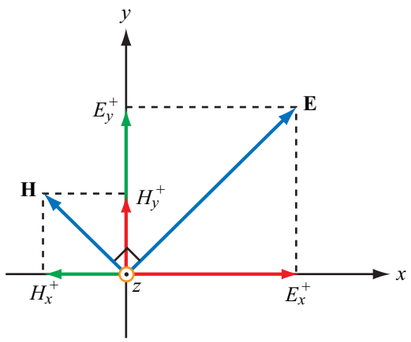


图 7-6 波 (E,H) 相当于两个波的和：一个波具有场 (E_x^+,H_y^+)，另一个波具有场 (E_y^+,H_x^+)，两者都沿 +z 方向传播。

概念问题 7-1：什么是均匀平面波？从物理和数学角度描述它的属性。在什么条件下适合将球面波视为平面波？

概念问题 7-2：由于 $\widetilde{\mathbf{E}}$ 和 $\widetilde{\mathbf{H}}$ 受相同形式的波动方程控制 [Eqs. (7.15) 和 (7.16)]，它是否遵循 $\widetilde{\mathbf{E}} = \widetilde{\mathbf{H}}$？解释。

概念问题 7-3：如果 TEM 波沿 $\hat{\mathbf{y}}$ 方向传播，其电场是否可以有沿 $\hat{\mathbf{x}}$、$\hat{\mathbf{y}}$ 和 $\hat{\mathbf{z}}$ 的分量？解释。

练习 7-1：10 MHz 均匀平面波在具有 $\mu = \mu_0$ 和 $\varepsilon_r =$ 的非磁性介质中传播 9. 求 (a) 相速度，(b) 波数，(c) 介质中的波长，以及 (d) 介质的固有阻抗。

Answer: (a) $u_p = 1 \times 10^8$ m/s, (b) $k = 0.2\pi$ rad/m, (c) $\lambda = 10$ m, (d) $\eta = 125.67\ \Omega$. (See EM.)

Exercise 7-2: The electric field phasor of a uniform plane wave traveling in a lossless medium with an intrinsic impedance of 188.5 Ω is given by $\widetilde{\mathbf{E}} = \hat{\mathbf{z}}\,10e^{-j4\pi y}$ (mV/m). Determine (a) the associated magnetic field phasor and (b) the instantaneous expression for $\mathbf{E}(y,t)$ if the medium is nonmagnetic ($\mu = \mu_0$).

Answer: (a) $\widetilde{\mathbf{H}} = \hat{\mathbf{x}}\,53e^{-j4\pi y}$ (μA/m), (b) $\mathbf{E}(y,t) = \hat{\mathbf{z}}\,10\cos(6\pi \times 10^8 t - 4\pi y)$ (mV/m). (See EM.)

Exercise 7-3: If the magnetic field phasor of a plane wave traveling in a medium with intrinsic impedance $\eta = 100\ \Omega$ is given by $\widetilde{\mathbf{H}} = (\hat{\mathbf{y}}\,10 + \hat{\mathbf{z}}\,20)e^{-j4x}$ (mA/m), find the associated electric field phasor.

Answer: $\widetilde{\mathbf{E}} = (-\hat{\mathbf{z}} + \hat{\mathbf{y}}\,2)e^{-j4x}$ (V/m). (See EM.)

Exercise 7-4: Repeat Exercise 7-3 for a magnetic field given by $\widetilde{\mathbf{H}} = \hat{\mathbf{y}}(10e^{-j3x} - 20e^{j3x})$ (mA/m).

Answer: $\widetilde{\mathbf{E}} = -\hat{\mathbf{z}}(e^{-j3x} + 2e^{j3x})$ (V/m). (See EM.)

7-3 Wave Polarization

▶ The ***polarization*** of a uniform plane wave describes the locus traced by the tip of the **E** vector (in the plane orthogonal to the direction of propagation) at a given point in space as a function of time. ◀

In the most general case, the locus of the tip of **E** is an ellipse, and the wave is said to be ***elliptically polarized***. Under certain conditions, the ellipse may degenerate into a circle or a straight line, in which case the ***polarization state*** is called ***circular*** or ***linear***, respectively.

It was shown in Section 7-2 that the z components of the electric and magnetic fields of a z-propagating plane wave are both zero. Hence, in the most general case, the electric field phasor $\widetilde{\mathbf{E}}(z)$ of a $+z$-propagating plane wave may consist of an x component, $\hat{\mathbf{x}}\,\widetilde{E}_x(z)$, and a y component, $\hat{\mathbf{y}}\,\widetilde{E}_y(z)$, or

$$\widetilde{\mathbf{E}}(z) = \hat{\mathbf{x}}\widetilde{E}_x(z) + \hat{\mathbf{y}}\widetilde{E}_y(z) \tag{7.46}$$

with

$$\widetilde{E}_x(z) = E_{x0}e^{-jkz}, \tag{7.47a}$$

$$\widetilde{E}_y(z) = E_{y0}e^{-jkz}, \tag{7.47b}$$

where E_{x0} and E_{y0} are the amplitudes of $\widetilde{E}_x(z)$ and $\widetilde{E}_y(z)$, respectively. For the sake of simplicity, the plus sign superscript has been suppressed; the negative sign in e^{-jkz} is sufficient to remind us that the wave is traveling in the positive z direction.

The two amplitudes E_{x0} and E_{y0} are, in general, complex quantities with each characterized by a magnitude and a phase angle. The phase of a wave is defined relative to a reference state, such as $z = 0$ and $t = 0$ or any other combination of z and t. As will become clear from the discussion that follows, the polarization of the wave described by Eqs. (7.46) and (7.47) depends on the phase of E_{y0} relative to that of E_{x0} but not on the absolute phases of E_{x0} and E_{y0}. Hence, for convenience, we assign E_{x0} a phase of zero and denote the phase of E_{y0}, relative to that of E_{x0}, as δ. Thus, δ is the *phase difference* between the y and x components of $\widetilde{\mathbf{E}}$. Accordingly, we define E_{x0} and E_{y0} as

$$E_{x0} = a_x, \tag{7.48a}$$

$$E_{y0} = a_y e^{j\delta}, \tag{7.48b}$$

where $a_x = |E_{x0}| \geq 0$ and $a_y = |E_{y0}| \geq 0$ are the magnitudes of E_{x0} and E_{y0}, respectively. Thus, by definition, a_x and a_y may not assume negative values. Using Eqs. (7.48a and b) in Eqs. (7.47a and b), the total electric field phasor is

$$\widetilde{\mathbf{E}}(z) = (\hat{\mathbf{x}}a_x + \hat{\mathbf{y}}a_y e^{j\delta})e^{-jkz}, \tag{7.49}$$

and the corresponding instantaneous field is

$$\begin{aligned}\mathbf{E}(z,t) &= \Re\mathfrak{e}\left[\widetilde{\mathbf{E}}(z)\, e^{j\omega t}\right] \\ &= \hat{\mathbf{x}}a_x\cos(\omega t - kz) + \hat{\mathbf{y}}a_y\cos(\omega t - kz + \delta).\end{aligned} \tag{7.50}$$

When characterizing an electric field at a given point in space, two of its attributes that are of particular interest are its magnitude and direction. The magnitude of $\mathbf{E}(z,t)$ is

$$\begin{aligned}|\mathbf{E}(z,t)| &= [E_x^2(z,t) + E_y^2(z,t)]^{1/2} \\ &= [a_x^2\cos^2(\omega t - kz) + a_y^2\cos^2(\omega t - kz + \delta)]^{1/2}.\end{aligned} \tag{7.51}$$

The electric field $\mathbf{E}(z,t)$ has components along the x and y directions. At a specific position z, the direction of $\mathbf{E}(z,t)$ is characterized by its ***inclination angle*** ψ, defined with respect to the x axis and given by

$$\psi(z,t) = \tan^{-1}\left[\frac{E_y(z,t)}{E_x(z,t)}\right]. \tag{7.52}$$

In the general case, both the intensity of $\mathbf{E}(z,t)$ and its direction are functions of z and t. Next, we examine some special cases.

答案：(a) $u_p = 1 \times 10^8$ m/s，(b) $k = 0.2\pi$ rad/m，(c) $\lambda = 10$ m，(d) $\eta = 125.67\ \Omega$。（参见EM。）

练习 7-2：在固有阻抗为 188.5 Ω 的无损介质中传播的均匀平面波的电场相量由 $\widetilde{E} = \hat{z}\ 10e^{-j4\pi y}$ (mV/m) 给出。如果介质是非磁性的 ($\mu = \mu_0$)，则确定 (a) 相关的磁场相量和 (b) E(y,t) 的瞬时表达式。

答案：(a) $\widetilde{H} = \hat{x}\ 53e^{-j4\pi y}$ (μ A/m)，(b) $E(y,t) = \hat{z}\ 10\cos(6\pi \times 10^8 t - 4\pi y)$ (mV/m)。（参见EM。）

练习 7-3：如果在固有阻抗为 $\eta = 100\ \Omega$ 的介质中传播的平面波的磁场相量由 $\widetilde{H} = (\hat{y}\ 10 + \hat{z}\ 20)e^{-j4x}$ (mA/m) 给出，求相关的电场相量。

Answer: $\widetilde{\mathbf{E}} = (-\hat{\mathbf{z}} + \hat{\mathbf{y}}\,2)e^{-j4x}$ (V/m). (See EM.)

练习 7-4：对由 $\widetilde{H} = \hat{y}(10e^{-j3x} - 20e^{j3x})$ (mA/m) 给出的磁场重复练习 7-3。

Answer: $\widetilde{\mathbf{E}} = -\hat{\mathbf{z}}(e^{-j3x} + 2e^{j3x})$ (V/m). (See EM.)

7-3 波的偏振

▶ 均匀平面波的偏振描述了 E 矢量尖端（在与传播方向正交的平面中）在空间中给定点处随时间变化的轨迹。 ◀

在最一般的情况下，E 尖端的轨迹是椭圆，并且该波被称为椭圆偏振。在某些条件下，椭圆可能退化为圆形或直线，在这种情况下，偏振态分别称为圆形或线性。

第 7-2 节表明，z 方向传播的平面波的电场和磁场的 z 分量均为零。因此，在最一般的情况下，+z 传播平面波的电场相量 $\widetilde{E}(z)$ 可能由 x 分量 $\hat{x}\ \widetilde{E}_x(z)$ 和 y 分量 $\hat{y}\ \widetilde{E}_y(z)$ 组成，或者

$$\widetilde{\mathbf{E}}(z) = \hat{\mathbf{x}}\widetilde{E}_x(z) + \hat{\mathbf{y}}\widetilde{E}_y(z) \tag{7.46}$$

和

$$\widetilde{E}_x(z) = E_{x0}e^{-jkz}, \tag{7.47a}$$

$$\widetilde{E}_y(z) = E_{y0}e^{-jkz}, \tag{7.47b}$$

其中E_{x0}和E_{y0}分别是$\widetilde{E}_x(z)$和$\widetilde{E}_y(z)$的幅度。为了简单起见，加号上标已被取消； e^{-jkz} 中的负号足以提醒我们波正向 z 方向传播。

通常，两个幅度 E_{x0} 和 E_{y0} 是复量，每个幅度都由幅度和相位角表征。波的相位是相对于参考状态定义的，例如 z = 0 和 t = 0 或 z 和 t 的任何其他组合。从下面的讨论中可以清楚地看出，方程描述的波的偏振。 (7.46) 和 (7.47) 取决于 E_{y0} 相对于 E_{x0} 的相位，但不取决于 E_{x0} 和 E_{y0} 的绝对相位。因此，为了方便起见，我们将 E_{x0} 的相位指定为零，并将 E_{y0} 相对于 E_{x0} 的相位表示为 δ。因此，δ 是 $\widetilde{E}$ 的 y 和 x 分量之间的相位差。因此，我们将 E_{x0} 和 E_{y0} 定义为

$$E_{x0} = a_x, \tag{7.48a}$$

$$E_{y0} = a_y e^{j\delta}, \tag{7.48b}$$

其中 $a_x = |E_{x0}| \geq 0$ 和 $a_y = |E_{y0}| \geq 0$ 分别是 E_{x0} 和 E_{y0} 的大小。因此，根据定义，a_x 和 a_y 不能取负值。使用方程式。等式中的（7.48a 和 b） （7.47a 和 b），总电场相量为

$$\widetilde{\mathbf{E}}(z) = (\hat{\mathbf{x}}a_x + \hat{\mathbf{y}}a_y e^{j\delta})e^{-jkz}, \tag{7.49}$$

相应的瞬时场为

$$\begin{aligned}\mathbf{E}(z,t) &= \Re\mathfrak{e}\left[\widetilde{\mathbf{E}}(z)\, e^{j\omega t}\right] \\ &= \hat{\mathbf{x}}a_x\cos(\omega t - kz) + \hat{\mathbf{y}}a_y\cos(\omega t - kz + \delta).\end{aligned} \tag{7.50}$$

当表征空间中给定点的电场时，特别令人感兴趣的两个属性是其大小和方向。 E(z,t) 的大小为

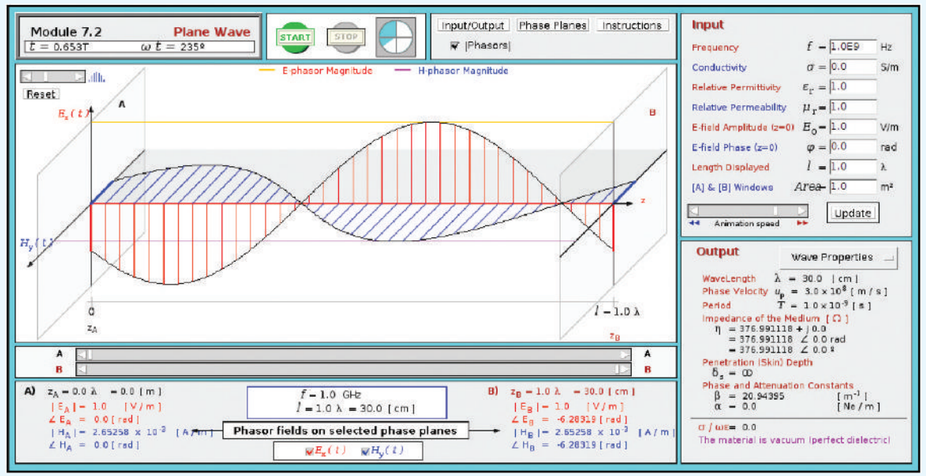
$$\begin{aligned}|\mathbf{E}(z,t)| &= [E_x^2(z,t) + E_y^2(z,t)]^{1/2} \\ &= [a_x^2\cos^2(\omega t - kz) + a_y^2\cos^2(\omega t - kz + \delta)]^{1/2}.\end{aligned} \tag{7.51}$$

电场 E(z,t) 具有沿 x 和 y 方向的分量。在特定位置 z，E(z,t) 的方向由其倾斜角 ψ 表征，相对于 x 轴定义并由下式给出

$$\psi(z,t) = \tan^{-1}\left[\frac{E_y(z,t)}{E_x(z,t)}\right]. \tag{7.52}$$

一般情况下，E(z,t) 的强度及其方向都是 z 和 t 的函数。接下来，我们研究一些特殊情况。

Module 7.2 Plane Wave Observe a plane wave propagating along the z direction; note the temporal and spatial variations of $\mathbf{E}$ and $\mathbf{H}$. Examine how the wave properties change as a function of the values selected for the wave parameters—frequency and $\mathbf{E}$ field amplitude and phase—and the medium's constitutive parameters $(\varepsilon, \mu, \sigma)$.



7-3.1 Linear Polarization

► A wave is said to be linearly polarized if for a fixed z, the tip of $\mathbf{E}(z,t)$ traces a straight line segment as a function of time. This happens when $E_x(z,t)$ and $E_y(z,t)$ are ***in phase*** (i.e., $\delta = 0$) or ***out of phase*** ($\delta = \pi$). ◄

Under these conditions, Eq. (7.50) simplifies to

$$\mathbf{E}(0,t) = (\hat{\mathbf{x}}a_x + \hat{\mathbf{y}}a_y)\cos(\omega t - kz), \quad \textbf{(in phase)} \tag{7.53a}$$

$$\mathbf{E}(0,t) = (\hat{\mathbf{x}}a_x - \hat{\mathbf{y}}a_y)\cos(\omega t - kz). \quad \textbf{(out of phase)} \tag{7.53b}$$

Let us examine the out-of-phase case. The field's magnitude is

$$|\mathbf{E}(z,t)| = [a_x^2 + a_y^2]^{1/2}|\cos(\omega t - kz)|, \tag{7.54a}$$

and the inclination angle is

$$\psi = \tan^{-1}\left(\frac{-a_y}{a_x}\right). \quad \textbf{(out of phase)} \tag{7.54b}$$

We note that ψ is independent of both z and t. **Figure 7-7** displays the line segment traced by the tip of $\mathbf{E}$ at $z = 0$ over a half of a cycle. The trace would be the same at any other value of z as well. At $z = 0$ and $t = 0$, $|E(0,0)| = [a_x^2 + a_y^2]^{1/2}$. The length of the vector representing $\mathbf{E}(0,t)$ decreases to zero at $\omega t = \pi/2$. The vector then reverses direction and increases in magnitude to $[a_x^2 + a_y^2]^{1/2}$ in the second quadrant of the x–y plane at $\omega t = \pi$. Since ψ is independent of both z and t, $\mathbf{E}(z,t)$ maintains a direction along the line making an angle ψ with the x axis while oscillating back and forth across the origin.

If $a_y = 0$, then $\psi = 0°$ or $180°$, and the wave is x-polarized; conversely, if $a_x = 0$, then $\psi = 90°$ or $-90°$, and the wave is y-polarized.

7-3.2 Circular Polarization

We now consider the special case when the magnitudes of the x and y components of $\widetilde{\mathbf{E}}(z)$ are equal, and the phase difference $\delta = \pm\pi/2$. For reasons that become evident shortly, the wave polarization is called ***left-hand circular*** when $\delta = \pi/2$, and ***right-hand circular*** when $\delta = -\pi/2$.

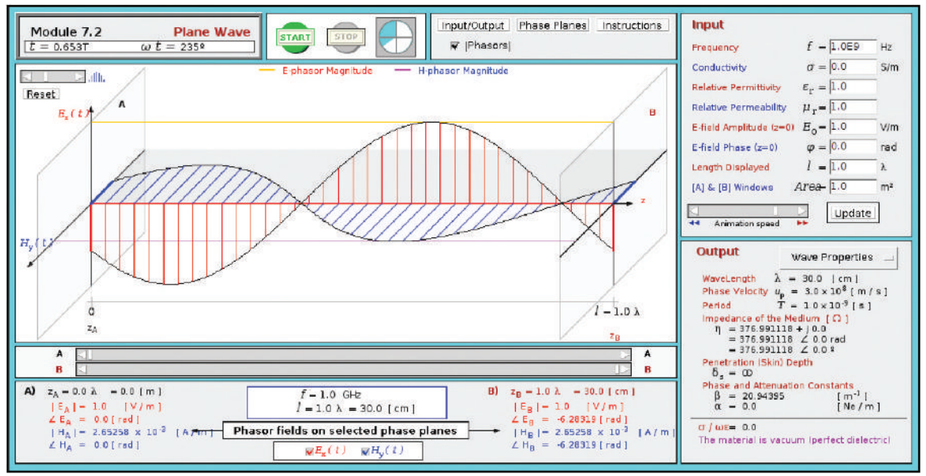
Left-Hand Circular (LHC) Polarization

For $a_x = a_y = a$ and $\delta = \pi/2$, Eqs. (7.49) and (7.50) become

$$\widetilde{\mathbf{E}}(z) = (\hat{\mathbf{x}}a + \hat{\mathbf{y}}ae^{j\pi/2})e^{-jkz} = a(\hat{\mathbf{x}} + j\hat{\mathbf{y}})e^{-jkz}, \tag{7.55a}$$

310 CHAPTER 7 PLANE-WAVE PROPAGATION

模块 7.2 平面波 观察沿 z 方向传播的平面波；注意 E 和 H 的时间和空间变化。检查波属性如何随着为波参数（频率和 E 场幅度和相位）以及介质的本构参数（ε、μ、σ）选择的值而变化。



7-3.1 线偏振

► 如果对于固定 z，E(z,t) 的尖端随时间描绘一条直线段，则称波是线性偏振的。当 $E_x(z,t)$ 和 $E_y(z,t)$ 同相（即 $\delta = 0$）或异相（$\delta = \pi$）时，就会发生这种情况。◄

在这些条件下，方程。(7.50) 简化为

$$\mathbf{E}(0,t) = (\hat{\mathbf{x}}a_x + \hat{\mathbf{y}}a_y)\cos(\omega t - kz), \quad \textbf{(in phase)} \tag{7.53a}$$

$$\mathbf{E}(0,t) = (\hat{\mathbf{x}}a_x - \hat{\mathbf{y}}a_y)\cos(\omega t - kz). \quad \textbf{(out of phase)} \tag{7.53b}$$

让我们检查一下异相情况。场的大小为

$$|\mathbf{E}(z,t)| = [a_x^2 + a_y^2]^{1/2}|\cos(\omega t - kz)|, \tag{7.54a}$$

倾斜角为

$$\psi = \tan^{-1}\left(\frac{-a_y}{a_x}\right). \quad \textbf{(out of phase)} \tag{7.54b}$$

我们注意到 ψ 独立于 z 和 t。图 7-7 显示了 E 尖端在 z = 0 处经过半个周期所描绘的线段。对于任何其他 z 值，轨迹也将相同。在 z = 0 和 t = 0 处，$|E(0,0)| = [a_x^2 + a_y^2]^{1/2}$。表示 E(0,t) 的向量长度在 $\omega t = \pi/2$ 处减小到零。然后，矢量反转方向并在 x-y 平面第二象限的 $\omega t = \pi$ 处增大幅度到 $[a_x^2 + a_y^2]^{1/2}$。由于 ψ 与 z 和 t 无关，因此 E(z,t) 保持沿与 x 轴形成角度 ψ 的线的方向，同时穿过原点来回振荡。

如果$a_y = 0$，则$\psi = 0°$或$180°$，并且波是x偏振的；反之，如果 $a_x = 0$，则 $\psi = 90°$ 或 $-90°$，且波为 y 偏振。

7-3.2 圆偏振

我们现在考虑当 $\widetilde{E}(z)$ 的 x 和 y 分量的幅度相等且相位差 $\delta = \pm\pi/2$ 时的特殊情况。由于很快就会显而易见的原因，当 $\delta = \pi/2$ 时，波偏振称为左旋圆偏振，而当 $\delta = -\pi/2$ 时，波偏振称为右旋圆偏振。

左手圆 (LHC) 偏振

对于 $a_x = a_y = a$ 和 $\delta = \pi/2$，等式(7.49) 和 (7.50) 变为

$$\widetilde{\mathbf{E}}(z) = (\hat{\mathbf{x}}a + \hat{\mathbf{y}}ae^{j\pi/2})e^{-jkz} = a(\hat{\mathbf{x}} + j\hat{\mathbf{y}})e^{-jkz}, \tag{7.55a}$$